

VegShift: Detecting Crop Viability Loss Events After Climate Zone Transitions in Indian Cities Using Multi-Source Geospatial and Temporal Deep Learning

AUTHOR NAME¹

¹Department, Institution, City, Country (e-mail: email@institution.ac.in)

Corresponding author: Author Name (e-mail: email@institution.ac.in).

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ABSTRACT Placeholder abstract: This paper presents VegShift, a framework for detecting Crop Viability Loss Events (CVLEs) triggered by Köppen-Geiger climate zone transitions across ten major Indian cities over 2000–2024, using a multi-source fusion of atmospheric, hydrological, and agro-ecological data with an eight-model comparative deep learning study.

INDEX TERMS crop viability loss, climate zone transition, Köppen-Geiger, groundwater depletion, temporal deep learning, TFT, CVLE, India agriculture

I. INTRODUCTION

CLIMATE-DRIVEN agricultural risk in India remains poorly quantified at the city scale, particularly the moment when a shifting climate permanently renders a crop non-viable. Existing work either uses static agroclimatic zones or single-variable thresholds; VegShift addresses this gap by formally timestamping viability loss events tied to climate zone transitions. This paper makes three contributions: (1) the CVLE dual-deficit event definition, (2) a 3-source geospatial dataset fusion spanning 25 years across 10 cities, and (3) a comparative study of 8 models with uncertainty quantification.

II. RELATED WORK

A. KÖPPEN-GEIGER CLASSIFICATION IN AGRICULTURE

Placeholder: Prior use of Köppen zones for agroclimatic zoning and its limitations under non-stationary climate.

B. CROP FAILURE PREDICTION

Placeholder: Survey of ML-based crop failure prediction methods, noting the predominance of single-variable rainfall or temperature thresholds.

C. GROUNDWATER DEPLETION AND AGRICULTURAL INTERACTION

Placeholder: Literature on CGWB-documented aquifer depletion trends and their link to irrigation-dependent crop failure in India.

D. TEMPORAL DEEP LEARNING FOR CLIMATE APPLICATIONS

Placeholder: Review of TFT, LSTM, TCN, and Transformer architectures applied to climate time-series classification and forecasting.

III. STUDY AREA AND DATA

A. STUDY CITIES AND CROP ASSIGNMENTS

Placeholder: Description of the 10 Indian cities (Delhi, Mumbai, Kolkata, Chennai, Bangalore, Hyderabad, Ahmedabad, Jaipur, Lucknow, Pune), their assigned crops, and the at-risk vs. control city grouping.

B. DATASET 1: DAILY ATMOSPHERIC CLIMATE

Placeholder: Open-Meteo / Kaggle Historical Weather Dataset covering 10 cities × 25 years of daily temperature, precipitation, and wind; humidity derived via Steadman apparent-temperature inversion.

C. DATASET 2: CGWB GROUNDWATER LEVELS

Placeholder: Central Groundwater Board seasonal well readings (2000–2022) for 2,759 wells aggregated within 50 km city radii; imputation strategy for Jaipur (nearest-well proxy) and post-2022 gap.

D. DATASET 3: FAO GAEZ V4 CROP SUITABILITY

Placeholder: GeoTIFF rasters at 9 km resolution providing rainfed baseline suitability scores (1–7) for each city-crop pair, extracted at city centroids.

IV. METHODOLOGY

A. KÖPPEN-GEIGER CLASSIFICATION AND TRANSITION DETECTION

Placeholder: Rule-based classification per Peel et al. (2007) applied annually; persistence filter requiring 3+ consecutive years before a transition is recorded.

B. ANNUAL FEATURE ENGINEERING

Placeholder: Derivation of 17 time-varying features (GDD accumulation, monsoon onset DOY, crop water deficit, recharge efficiency, dual-deficit flag, etc.) and 5 static features from ECOCROP and GAEZ.

C. CROP VIABILITY LOSS EVENT (CVLE) DEFINITION

Placeholder: Formal dual-deficit definition requiring consecutive-year atmospheric and hydrological failure with at least 2 of 3 threshold breaches; pseudocode provided as Algorithm ??.

Algorithm 1 CVLE Label Construction

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1: for each city do
2:   for  $t = 1$  to  $T$  do
3:      $consec \leftarrow dual\_deficit[t] = 1 \wedge dual\_deficit[t-1] = 1$ 
4:      $s \leftarrow \mathbb{I}[sowing\_window\_miss[t] > 0.6] + \mathbb{I}[crop\_water\_deficit[t] > 0.4] + \mathbb{I}[gdd\_adequate[t] = 0]$ 
5:      $cvle\_label[t] \leftarrow 1$  if  $consec \wedge s \geq 2$  else 0
6:   end for
7: end for

```

D. MODEL ARCHITECTURES

Placeholder: Overview of all 8 models — Temporal Fusion Transformer (TFT), Random Forest, Logistic Regression, XGBoost, LightGBM, LSTM, TCN, and Vanilla Transformer — with hyperparameters.

E. TRAIN/VALIDATION/TEST SPLIT

Placeholder: Temporal split (train 2000–2018, val 2000–2021, test 2022–2024) with no data shuffling to prevent leakage; class imbalance (7.6% positive) and handling strategies.

F. EVALUATION METRICS AND THRESHOLD OPTIMISATION

Placeholder: AUC-ROC, macro-F1, Brier score, accuracy; classification threshold selected by F1-macro grid search over the validation set rather than fixed at 0.5.

G. UNCERTAINTY QUANTIFICATION

Placeholder: Monte Carlo dropout (50 forward passes) for LSTM, TCN, and Transformer; tree variance for Random Forest; Expected Calibration Error (ECE) computed with 10-bin histogram.

H. FEATURE GROUP ABLATION

Placeholder: Five mutually exclusive feature groups (Climate, Phenology, Hydrology, Static Context, All Features) evaluated independently on RF, XGB, and LSTM.

V. EXPERIMENTS AND RESULTS

A. UNIFIED TEST-SET PERFORMANCE

Placeholder: Table comparing all 8 models on AUC-ROC, macro-F1, Brier score, accuracy, and optimal threshold (test set: $n = 30$, years 2022–2024).

B. PAIRWISE STATISTICAL SIGNIFICANCE

Placeholder: Wilcoxon signed-rank test on per-sample Brier loss across 28 model pairs; results indicate which performance differences are statistically significant at $p < 0.05$.

C. FEATURE GROUP ABLATION RESULTS

Placeholder: Static context alone (GAEZ baseline + Köppen encoding) achieves $AUC \geq 0.98$ for tree models; hydrological features are the dominant group for recurrent models.

D. UNCERTAINTY QUANTIFICATION RESULTS

Placeholder: Random Forest tree-variance achieves the lowest ECE (0.045), outperforming all MC-dropout neural methods; interval widths and mean prediction std reported per model.

E. SHAP FEATURE IMPORTANCE

Placeholder: Global mean |SHAP| rankings from RF, XGB, and LGB; `gaez_baseline_class` and `koppen_zone_enc` dominate, confirming ablation findings.

F. CVLE EVENT TIMELINE PER CITY

Placeholder: Dated CVLE events for the 7 at-risk cities; control cities (Pune, Kolkata, Mumbai) produce near-zero CVLE rates, validating the dual-deficit formulation.

VI. DISCUSSION

A. WHY SIMPLE MODELS OUTPERFORM DEEP MODELS

Placeholder: The CVLE classification problem is largely linearly separable in feature space at $n = 250$; LR achieves $AUC = 1.0$ while TFT collapses to $AUC = 0.069$.

B. WHY TFT FAILS

Placeholder: QuantileLoss is designed for multi-horizon regression, not binary classification; the 250-row dataset is insufficient for the 115k-parameter model.

C. ROLE OF STATIC VERSUS DYNAMIC FEATURES

Placeholder: Static GAEZ and Köppen features carry near-complete predictive signal; time-varying features add marginal value except for recurrent models exploiting groundwater trends.

D. GROUNDWATER AS THE DIFFERENTIATING SIGNAL FOR RECURRENT MODELS

Placeholder: Hydrological features achieve LSTM AUC=0.8621, the highest single-group AUC for sequential architectures, underscoring aquifer depletion as a lagged viability signal.

E. LIMITATIONS AND VALIDITY THREATS

Placeholder: Key limitations include the small test set ($n = 30$, 1 positive), CGWB data ending in 2022, Jaipur nearest-well proxy, humidity estimation from temperature inversion, and subjectively chosen CVLE thresholds.

VII. CONCLUSION

VegShift introduces the first formal dual-deficit Crop Viability Loss Event framework tied to Köppen climate zone transitions, validated across 10 Indian cities over 25 years. Random Forest is the recommended deployment model (AUC=0.9655, ECE=0.045), while the CVLE definition is validated by near-zero false detection in stable control cities. Future work should extend to a wider geographic scope and incorporate ground-truth crop yield records for threshold calibration.

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REFERENCES

- [1] Placeholder reference.

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